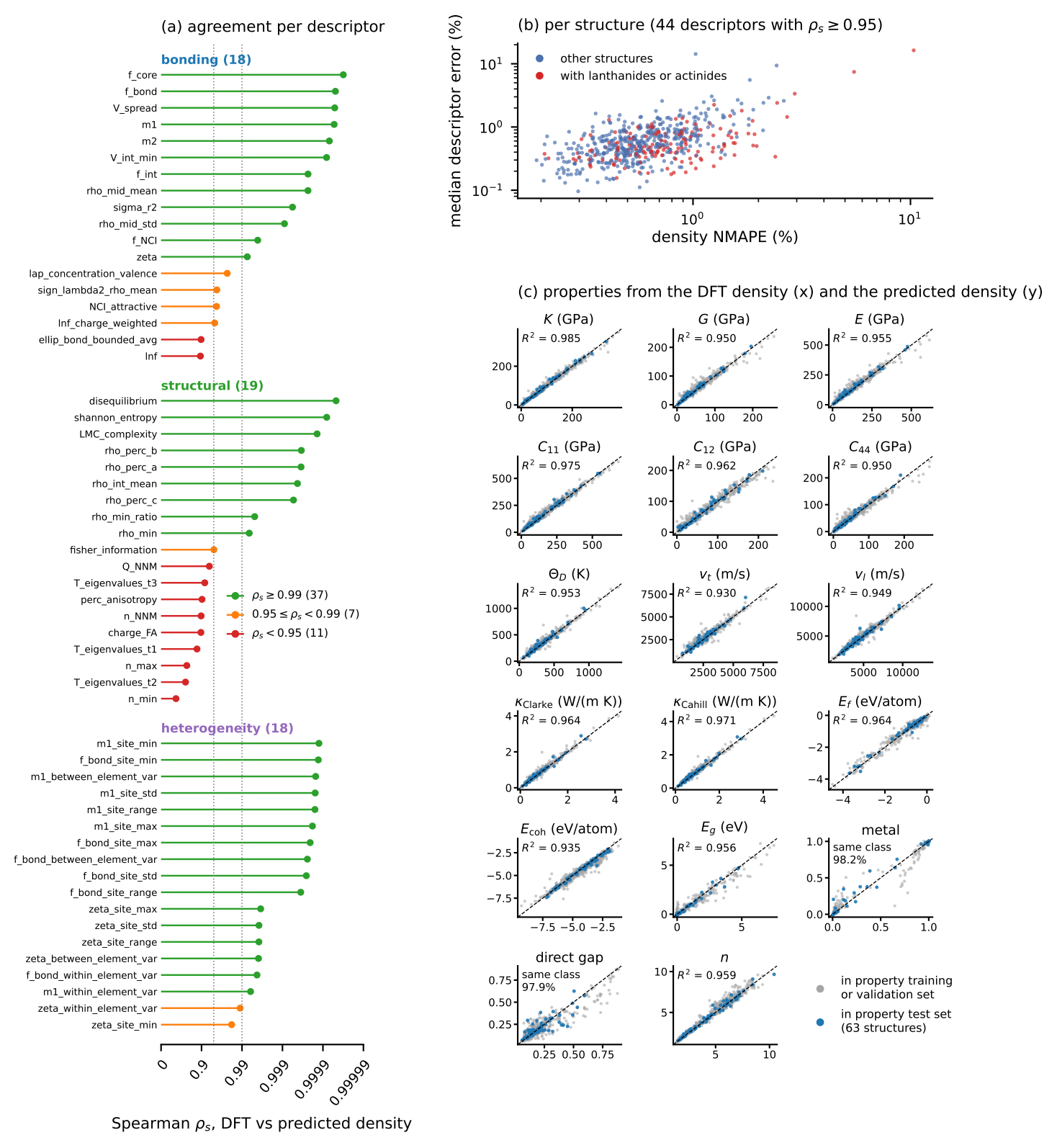

(a) agreement per descriptor
bonding (18)
f_core
f_bond
V_spread
m1
m2
V_int_min
f_int
rho_mid_mean
sigma_r2
rho_mid_std
f_NCI
zeta
lap_concentration_valence
sign_lambda2_rho_mean
NCI_attractive
lnf_charge_weighted
ellip_bond_bounded_avg
lnf
structural (19)
disequilibrium
shannon_entropy
LMC_complexity
rho_perc_b
rho_perc_a
rho_int_mean
rho_perc_c
rho_min_ratio
rho_min
fisher_information
Q_NNM
T_eigenvalues_t3
perc_anisotropy
n_NNM
charge_FA
T_eigenvalues_t1
n_max
T_eigenvalues_t2
n_min
heterogeneity (18)
m1_site_min
f_bond_site_min
m1_between_element_var
m1_site_std
m1_site_range
m1_site_max
f_bond_site_max
f_bond_between_element_var
f_bond_site_std
f_bond_site_range
zeta_site_max
zeta_site_std
zeta_site_range
zeta_between_element_var
f_bond_within_element_var
m1_within_element_var
zeta_within_element_var
zeta_site_min
$\rho_s \geq 0.99$ (37)
$0.95 \leq \rho_s < 0.99$ (7)
$\rho_s < 0.95$ (11)
0
0.9
0.99
0.999
0.9999
0.99999
Spearman ρ_s, DFT vs predicted density
(b) per structure (44 descriptors with $\rho_s \geq 0.95$)
other structures
with lanthanides or actinides
median descriptor error (%)
density NMAPE (%)
(c) properties from the DFT density (x) and the predicted density (y)
K (GPa)
R^2 = 0.985
G (GPa)
R^2 = 0.950
E (GPa)
R^2 = 0.955
C_{11} (GPa)
R^2 = 0.975
C_{12} (GPa)
R^2 = 0.962
C_{44} (GPa)
R^2 = 0.950
Θ_D (K)
R^2 = 0.953
v_t (m/s)
R^2 = 0.930
v_l (m/s)
R^2 = 0.949
κ_{Clarke} (W/(m K))
R^2 = 0.964
κ_{Cahill} (W/(m K))
R^2 = 0.971
E_f (eV/atom)
R^2 = 0.964
E_{coh} (eV/atom)
R^2 = 0.935
E_g (eV)
R^2 = 0.956
metal
same class
98.2%
direct gap
same class
97.9%
n
R^2 = 0.959
in property training or validation set
in property test set (63 structures)